

Explainable AI Report

Introduction

Explainable Artificial Intelligence (XAI) improves the transparency of deep learning models by providing visual explanations for their predictions. In this project, **Grad-CAM (Gradient-weighted Class Activation Mapping)** was used to highlight the regions of skin lesion images that contributed most to the prediction made by the **ResNet50 (SGD)** model.

Explainable AI Technique

Grad-CAM

Grad-CAM generates a heatmap by using the gradients of the predicted class with respect to the last convolutional layer. The heatmap highlights the most influential regions of the input image, helping to understand how the model arrived at its prediction.

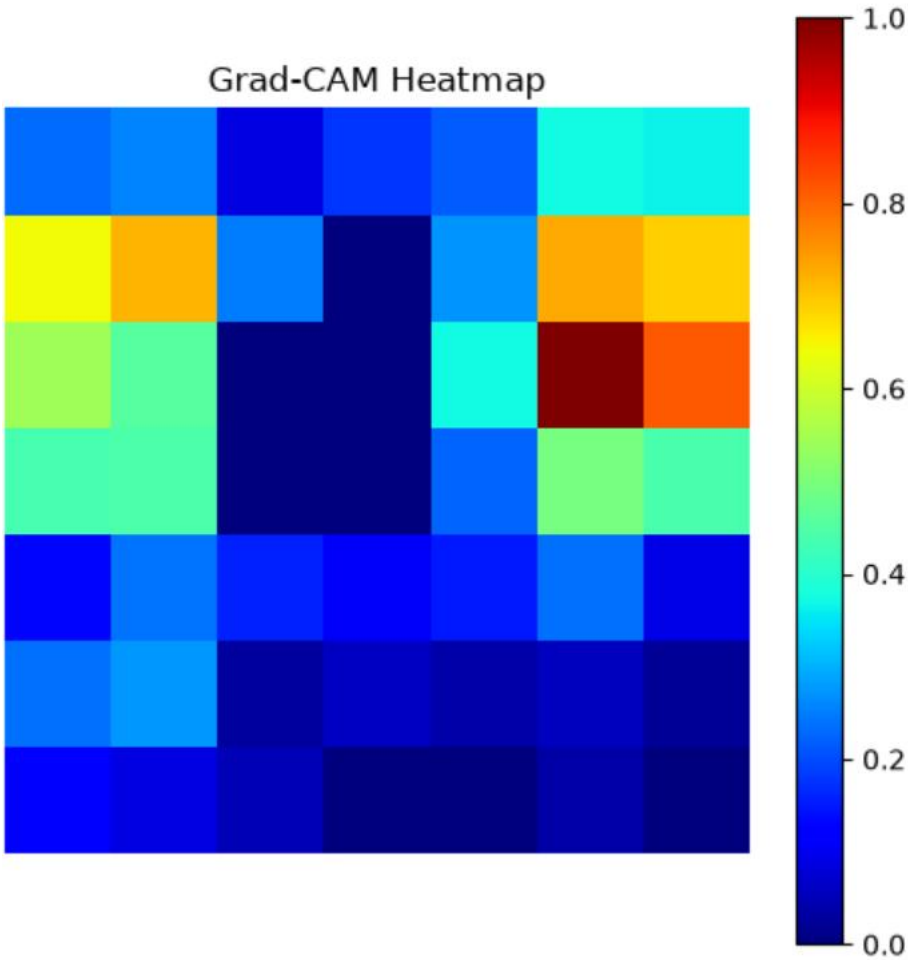
Grad-CAM Results

The Grad-CAM visualization was generated for the predicted skin lesion image. The heatmap clearly highlights the lesion region that contributed most to the model's decision. An overlay image was also created by combining the heatmap with the original image, providing a more intuitive visualization of the model's attention.

Original Skin Leison Image

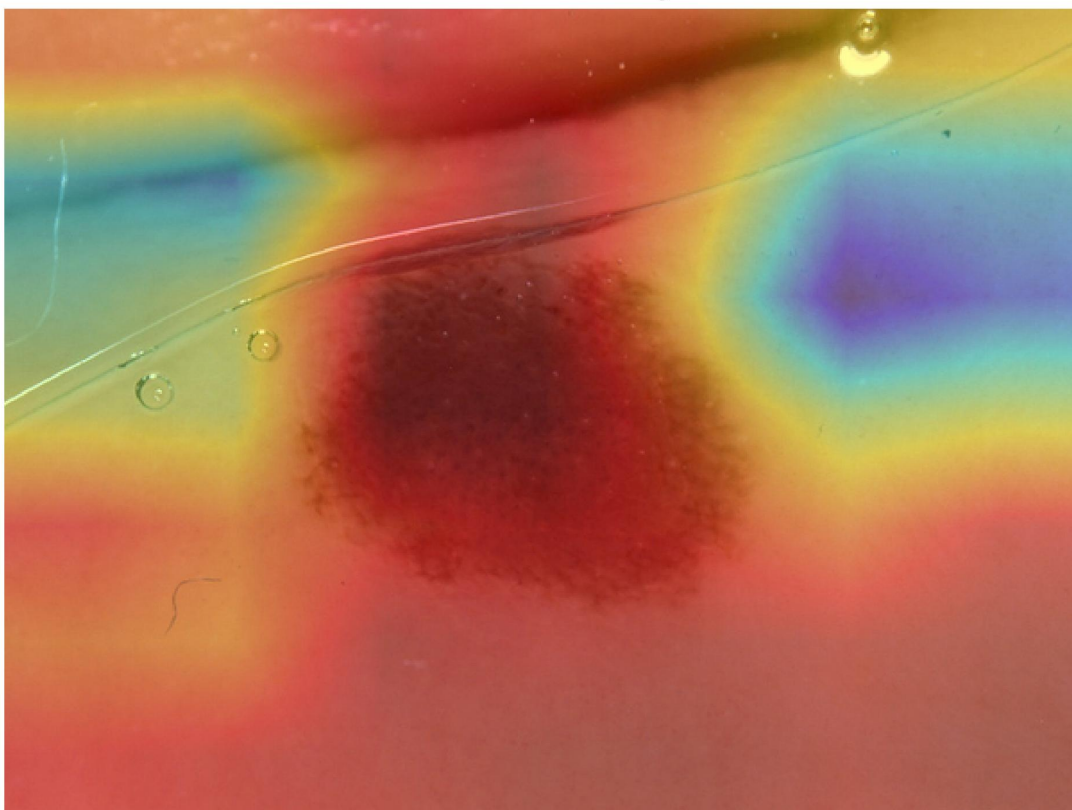


Grad-CAM Heatmap



Grad-CAM Overlay

Grad-CAM Overlay

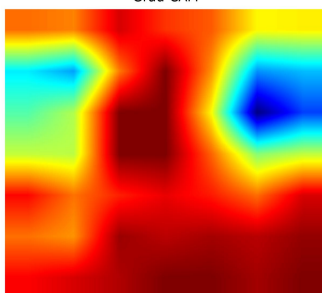


Combined Visualization

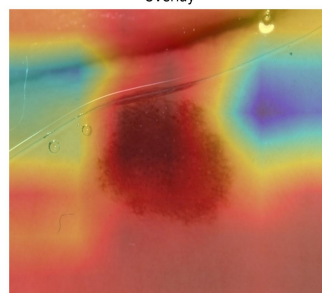
Original Image



Grad-CAM



Overlay



Summary

The Grad-CAM results demonstrate that the **ResNet50 (SGD)** model focuses on the relevant lesion regions while making predictions. These visual explanations improve the interpretability of the model and increase confidence in its classification results, making the model more reliable for deployment in the skin disease diagnosis application.